Dual-Fault Tolerant Control for Connected Vehicular Platoons Based on Extended State Fuzzy Observation

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***Abstract*—This article investigates an adaptive fuzzy active fault-tolerant control problem for connected automated vehicular platoons operating under actuator faults and inter-vehicle ranging sensor faults. A saturated Nussbaum function is employed to handle actuator faults, suppressing the initial acceleration oscillations caused by conventional Nussbaum-based control and thereby improving the transient response of the vehicular platoon. Simultaneously, within the control framework, an adaptive compensation scheme is constructed to restore the reliability of inter-vehicle distance measurements compromised by onboard ranging sensor faults. In addition, an extended state fuzzy observer is designed to estimate both the system states and external disturbances, thereby overcoming the difficulty of obtaining reliable and accurate vehicle motion information in the presence of internal dynamic uncertainties and external disturbances. Finally, the stability of the vehicular platoon can be verified by Lyapunov stability criteria, and sufficient validations on eight vehicles are implemented for validating the efficacy of the proposed control approach.**

***Impact Statement*—Reliable operation of vehicular platoons in the presence of component faults is a key obstacle to their deployment. This work shows that it is possible to maintain stable spacing and smooth longitudinal motion for a third-order nonlinear platoon even when actuator faults occur and onboard ranging sensors provide faulty distance measurements. By adaptively adjusting the control input under faults, correcting faulty inter-vehicle distance measurements, and estimating external disturbances online, the proposed framework achieves internal stability and string stability of the vehicular platoon. These capabilities enhance the resilience and practical applicability of platoon control schemes under realistic fault conditions.**

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***Index Terms*—Actuator and sensor faults, fault-tolerant control, fuzzy logic system, vehicular platoons.**

I. Introduction

DRIVEN by rapid economic growth and technological advancements, the exponential rise in private vehicle ownership has fueled severe urban traffic congestion, emerging as a pressing social concern. However, given the limitations of natural and urban resources, expanding road infrastructure to mitigate congestion is often infeasible [1], [2], [3]. Recent studies indicate that connected automated vehicles (CAVs) offer a promising solution to this challenge [4], [5]. In particular, vehicular platoons, as a series of CAVs moving straight along a road, have shown great potential in reducing emissions, improving traffic throughput, and enhancing road safety [6], [7]. Consequently, research on vehicular platoons has surged, yielding significant advances in control strategies [8], communication and privacy [9], and system security [10]. Various advanced control approaches, including backstepping control [11], sliding mode control [12], and model predictive control [13], have been applied to enhance the performance of individual vehicles within platoons.

While extensive research has explored advanced control strategies for vehicular platoons, achieving and maintaining key performance indicators, such as internal stability and string stability, remains a fundamental objective. Building on this foundation, further research has focused on improving transient performance to achieve improved convergence speed and limited tracking errors [14], [15], [16], [17]. To achieve these requirements, the authors in [18] proposed a finite-time sliding-mode controller that guaranteed both internal and string stability within a finite horizon. In [19], an adaptive fuzzy prescribed-time controller was proposed to ensure that the followers track the leader's velocity within a prescribed time. The tracking problem of third-order nonlinear vehicular platoons with actuator constraints was investigated in [20]. However, the aforementioned studies overlook actuator and sensor faults, which would compromise the feasibility of the above strategies in practical scenarios. Consequently, fault-tolerant control for vehicular platoons has emerged as a pivotal research topic with substantial practical significance.

Actuator and sensor faults are unavoidable in real-world vehicular platoons due to component aging and prolonged use.

Typical examples include stuck, float, or loss-of-effectiveness faults in sensors and actuators. Even a single such fault may disrupt the coordination of the entire platoon and potentially lead to system-wide paralysis [21], [22], [23]. To address potential faults in vehicular platoons, fault diagnosis (FD) mechanisms have been extensively investigated. In [24], a network-based actuator fault-detection filter with a fault-weighting scheme was developed to enhance detection performance. For third-order vehicular platoons, [25] developed a high-gain-observer-based diagnostic scheme and a distributed data-driven fault-tolerant controller to handle faulty sensor signals. In [26], a safety consensus strategy combining fault detection with a distributed finite-time fault-tolerant controller was proposed to address sensor and actuator faults in vehicular platoons. Although some existing works have focused on the active fault-tolerant control (AFTC) with FD mechanisms, two critical limitations persist. First, FD in complex systems may suffer from inaccuracies, such as false alarms and missed detections. Second, excessive delays in fault response can lead to catastrophic and irreversible consequences for system stability [27]. Therefore, it is essential to develop fault-tolerant control strategies that can guarantee vehicular platoon stability and performance independently of FD mechanisms.

To avoid the aforementioned issues, AFTC schemes that do not rely on FD have been extensively investigated. By dynamically adjusting controller parameters, these approaches attenuate the impact of faults while preserving system performance, thereby eliminating the need for FD mechanisms [28]. In [29], an optimal active fuzzy fault-tolerant control scheme was developed for a class of multi-input and multi-output nonlinear systems with unknown perturbations, which can automatically accommodate nonlinear actuator and sensor faults. In [30], an adaptive fault-tolerant control scheme based on control gain reconfiguration was proposed to address actuator faults in a spacecraft system. In [31], a novel extended state observer was employed to estimate the system states and lumped disturbances, including external disturbances and actuator faults. In particular, recent engineered solutions for vehicular platoons subject to actuator and sensor faults increasingly adopt adaptive AFTC architectures.

Several adaptive AFTC based designs have been proposed for vehicular platoons [32], [33], [34], [35], [36], [37]. In [32], a neural-network-based approach was employed to estimate faulty sensor signals for fault-tolerant control. In [33], [34], [35], [36], AFTC strategies were designed for vehicular platoons with actuator faults. In [37], for heterogeneous uncertain nonlinear second-order autonomous vehicular platoons, an adaptive mechanism was embedded in a fault-tolerant controller to enhance the vehicles' resilience to faults. However, the aforementioned studies predominantly address a single fault type and focus on faults at the level of individual vehicle dynamics. Simultaneous actuator and sensor faults in vehicular platoons have received relatively limited attention. Meanwhile, in practical applications such as cooperative adaptive cruise control [38], onboard ranging sensors (e.g., radar) provide the relative distance measurements that are crucial for maintaining proper spacing and ensuring string stability within the vehicular platoon. Nevertheless, most existing AFTC works implicitly assume that these inter-vehicle distance measurements are reliable and design controllers to handle actuator faults or faults in ego-vehicle state sensors. In contrast, faults occurring in the inter-vehicle sensing channel, which render the true inter-vehicle distance unavailable, have rarely been explicitly addressed in the existing AFTC literature [39].

On the basis of the aforementioned discussion, these limitations motivate the adaptive fuzzy AFTC scheme proposed in this article, which simultaneously addresses faults in both the individual vehicle dynamics and the inter-vehicle sensing layer, while guaranteeing internal stability and string stability of vehicular platoons. The key contributions are as follows.

1) In vehicular platoons, conventional Nussbaum-based fault-tolerant controllers [40], [41] can induce pronounced acceleration oscillations under actuator loss-of-effectiveness faults, which limits their practical applicability. To address this issue, a saturated Nussbaum function is adopted within the AFTC framework, which alleviates acceleration oscillations while retaining the ability to cope with unknown fault gains, thereby enhancing the practical feasibility of the proposed AFTC strategy for vehicular platoons.
2) Unlike most existing AFTC studies on vehicular platoons, which typically assume reliable inter-vehicle distance measurements and primarily address faults in ego-vehicle state sensors [25], [26], [42], this work accounts for faults in onboard ranging sensors that distort the true inter-vehicle distance. An adaptive compensation mechanism is developed in the AFTC architecture to correct faulty distance measurements, thereby preserving accurate sensed spacing information for platoon control.
3) Different from most existing extended state observers [43], [44] that lump unknown nonlinear dynamics and external disturbances into a total disturbance, this article develops an extended state fuzzy observer that separates the estimation of nonlinear dynamics from that of external disturbances. The unknown nonlinear dynamics are approximated by fuzzy logic systems, while the external disturbance is reconstructed through augmented states, thereby providing a more structured and targeted estimation architecture.

The chapters of this article are as follows. Section II formulates the system dynamics and the control problem. Section III develops an extended state observer-based fuzzy fault-tolerant controller within a backstepping framework. Section IV provides the stability analysis of the vehicular platoon. Section V provides numerical simulations of the proposed strategy. Section VI concludes the work.

II. Preliminaries

As illustrated in Fig. 1, this article considers a vehicular platoon composed of one leader and n follower vehicles. Each follower is equipped with an onboard ranging sensor that measures the inter-vehicle distance to its immediate predecessor, given by $y_{1,i} = x_{1,i-1} - x_{1,i} - l_{i-1}$, where l_i denotes the length of

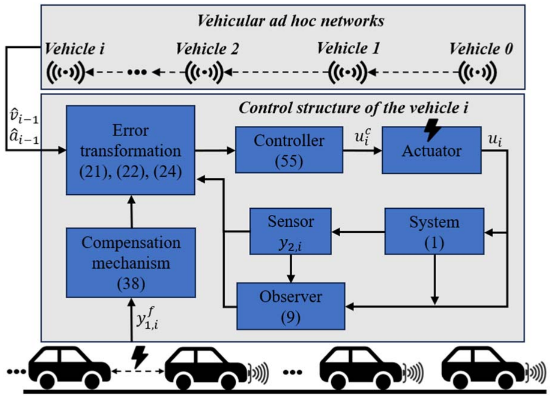

Fig. 1. Communication topology and control structure in vehicular platoon.

the ith vehicle. A predecessor-following (PF) communication topology with reliable communication channels is considered. In this topology, each follower receives information from its immediate predecessor via vehicular ad hoc networks and the leader influences the platoon indirectly.

The vehicular platoon operates under a prescribed constant distance (CD) policy, in which the desired headway between adjacent vehicles remains fixed. The position, velocity, and acceleration of the ith vehicle are denoted as $x_{1,i}$, $x_{2,i}$, and $x_{3,i}$, respectively, where $i \in \{0, 1, \ldots, n\}$ and $i = 0$ corresponds to the leader.

A. Vehicular Dynamics

The longitudinal dynamics of the ith follower are modeled by the following third-order system [45], [46]:

$$\begin{cases} \dot{x}_{1,i} = x_{2,i} + h_{1,i}\left(\bar{x}_{1,i}\right) \\ \dot{x}_{2,i} = x_{3,i} + h_{2,i}\left(\bar{x}_{2,i}\right) \\ \dot{x}_{3,i} = \frac{u_i}{m_i \tau_i} + h_{3,i}\left(\bar{x}_{3,i}\right) + D_i \\ y_{2,i} = x_{1,i} \end{cases} \quad (1)$$

where variables $x_{1,i}$, $x_{2,i}$ and $x_{3,i}$ denote the position, velocity and acceleration of the ith follower vehicle, respectively, and $\bar{x}_{k,i} = [x_{1,i}, \ldots, x_{k,i}]^{\mathrm{T}}$ collects the first k states. The input u_i is the throttle/brake signal, and $y_{2,i} = x_{1,i}$ is the measured longitudinal displacement output. The functions $h_{k,i}(\cdot)$ $(k = 1, 2, 3)$ represent unknown internal nonlinear dynamics. In particular, $h_{3,i}(\bar{x}_{3,i}) = -(k_{d,i}/m_i\tau_i)(x_{2,i}^2 + 2\tau_i x_{2,i} x_{3,i}) - (x_{3,i}/\tau_i)$, where m_i is the vehicle mass, τ_i is the engine time constant, and $k_{d,i}$ is the aerodynamic drag coefficient. The term D_i denotes external disturbances such as road slope.

Remark 1: In contrast to the models presented in [47], [48], the nonlinear third-order vehicular dynamics considered in this article provide a more general description of the vehicle's nonlinear properties, thus better reflecting real-world behavior. In practice, the required states are obtained from independent sensing modules: the relative position is measured by radar, while the ego-vehicle velocity and acceleration are provided by a speedometer and an accelerometer, respectively [39]. The nonlinearities introduced by these sensing modules are denoted by the unknown functions $h_{1,i}$ and $h_{2,i}$.

In particular, the dynamics of the leader vehicle are modeled as follows [49]:

$$\begin{cases} \dot{x}_{1,0} = x_{2,0} \\ \dot{x}_{2,0} = x_{3,0} \\ \dot{x}_{3,0} = \frac{u_0}{m_0 \tau_0} \end{cases} \quad (2)$$

where u_0 is the corresponding throttle/brake input variable.

To facilitate the design of an adaptive fuzzy AFTC, the following common assumptions and definitions are introduced.

Assumption 1 [50]: Suppose that the function $h_{k,i}$ satisfies the Lipschitz condition: for $\forall x_1, x_2$, there is a known constant $r_{k,i}$ such that

$$h_{k,i}\left(x_1\right) - h_{k,i}\left(x_2\right) \leq r_{k,i}\|x_1 - x_2\|.$$

Assumption 2 [51]: The unknown external disturbance D_i, its time derivative $\dot{D}_i$, and the bias fault u_{di} are all bounded.

Definition 1 (String stability) [52]: For any $\varrho > 0$, there exists a $\iota > 0$ such that, if the initial spacing errors satisfy $\max_i |e_{1,i}(0)| < \iota$, then $\sup_{t \geq 0} \max_i |e_{1,i}(t)| < \varrho$.

Definition 2 (Internal stability) [20]: Each vehicle maintains a safe distance from preceding vehicles and follows a reference trajectory in the aspects of speed and acceleration.

Definition 3 (Traffic flow stability) [43]: The vehicular platoon is regarded as traffic flow stable if $(\partial f / \partial \rho) > 0$, where f and ρ represent the traffic flow rate and density, respectively.

Remark 2: The Lipschitz condition in Assumption 1 is a standard requirement in nonlinear control that guarantees existence and uniqueness of solutions and facilitates Lyapunov stability analysis. In longitudinal vehicle dynamics, the disturbance mainly arises from bounded environmental effects such as aerodynamic drag and rolling resistance, while the bias fault represents a bounded offset in the throttle/brake channel. Therefore, Assumptions 1 and 2 are reasonable.

The goals of this article are as follows.

1) Achieve internal stability, string stability and traffic flow stability of the vehicular platoon.
2) Ensure that all signals in the vehicular platoon maintain boundedness under the proposed AFTC.

B. Fault Model

In practice, various types of faults may occur within a vehicular platoon, potentially affecting key actuators such as the throttle or brake due to unpredictable events. Such faults can lead to paralysis of the entire platoon, posing significant safety risks to vehicle operation. The actuator faults are modeled as follows [26]:

$$u_i = g_{0,i} u_i^c + u_{di} \quad (3)$$

where u_i^c is the actual control signal, $g_{0,i}$ are the unknown actuator effectiveness factor, and it meets $0 < g_{0,i} \leq 1$, and u_{di} represents an unknown bounded bias fault.

The model $y_{1,i}$ is established under the assumption that the true relative position can be measured by onboard ranging sensors, primarily radar. However, in practical scenarios, radar sensors may experience faults that cause deviations from their intended functionality [53]. Therefore, the general fault model presented in [22] is adopted to characterize faults in the relative distance sensor of the ith vehicle.

$$y_{1,i}^{f} = g_{1,i} y_{1,i} \tag{4}$$

where $0 < g_{1,i} \le 1$ denotes the effectiveness factor of the ith follower's onboard ranging sensor.

Remark 3: Equation (4) depicts a partial loss of effectiveness in the measurement channel, which may occur when onboard ranging sensors, such as radar or lidar, suffer from performance degradation. Such degradation can result from harsh environmental conditions, component aging, electromagnetic interference, leading to inaccurate distance measurements.

Remark 4: This article considers loss-of-effectiveness faults denoted by $g_{k,i}$ for both actuators and sensors. In the case of $g_{k,i} = 1$, components operate normally. For $0 < g_{k,i} < 1$, partial effectiveness loss faults occur. Specifically, when $g_{k,i} = 0$, the corresponding sensor or actuator produces zero output, leading to a total loss of effectiveness and rendering the vehicular platoon uncontrollable. Hence, this case is excluded from consideration.

C. Saturated Nussbaum Function

To alleviate actuator loss-of-effectiveness faults while avoiding the undesirable transient acceleration behaviors induced by conventional Nussbaum functions (e.g., $\cos(\pi\varsigma)e^{\varsigma^2}$, $\cos(\varsigma)\varsigma^2$), this article adopts the saturated Nussbaum function proposed in [54], defined as follows:

$$N(\varsigma) = \begin{cases} (-1)^{n+1} a \sin\left[b^n\left(\varsigma + \varsigma_{n-1}\right)\right], \\ \qquad \varsigma \in (-\varsigma_n, -\varsigma_{n-1}],\ n = 1, 2, \ldots, \\ a\sin(\varsigma), \qquad \varsigma \in (-\pi, \pi], \\ (-1)^n a \sin\left[b^n\left(\varsigma - \varsigma_{n-1}\right)\right], \\ \qquad \varsigma \in (\varsigma_{n-1}, \varsigma_n],\ n = 1, 2, \ldots. \end{cases}$$

where $a > 1, 0 < b < 1$, and $\varsigma_{n-1} = \sum_{k=0}^{n-1}(\pi/b^k)$, $\varsigma_n = \sum_{k=0}^{n}(\pi/b^k)$.

Lemma 1 [55]: Suppose $V(t) \ge 0$ and $\varsigma_m(t)$ are smooth functions on $[0, \infty)$, and $N(\cdot)$ is a smooth Nussbaum function. If there are positive constants C, D, and nonzero constants g_m such that

$$\dot{V} \le -CV + D + \sum_{m=1}^{n}\left(g_m N\left(\varsigma_m\right) + 1\right)\dot{\varsigma}_m$$

then $V(t)$ and $\varsigma_m(t)$ and $\sum_{m=1}^{n}\left(g_m N\left(\varsigma_m\right) + 1\right)\dot{\varsigma}_m$ are bounded on $[0, \infty)$.

D. Fuzzy Logic System

Lemma 2 [56]: Any continuous nonlinear function defined over a compact set Ω_i can be approximated via a fuzzy logic system as follows:

$$f_i(z_i) = \theta_i^{*T}\phi_i(z_i) + \epsilon_i \tag{5}$$

$$f_i(z_i) = \hat{\theta}_i^{T}\phi_i(z_i) + \nu_i \tag{6}$$

where $z_i = [z_{1,i}, \ldots, z_{j,i}]^T$ denotes the input vector to the fuzzy logic system. Specifically, $\hat{f}_i(z_i) = \hat{\theta}_i^T\phi_i(z_i)$ denotes the practical fuzzy approximation. The terms ϵ_i and ν_i represent the minimum approximation error and the approximation error, with $\nu_i := f_i(z_i) - \hat{f}_i(z_i)$, respectively. $\hat{\theta}_i$ is the estimated weight vector and the optimal weight vector θ_i^* is defined as follows:

$$\theta_i^* = \arg\min_{\theta_i \in \Pi_i}\ \sup_{z_i \in \Omega_i}\left|\theta_i^T\phi_i(z_i) - f_i(z_i)\right| \tag{7}$$

where $\theta_i \in \Pi_i$ is a candidate weight vector and Ω_i is a compact region for z_i. The weight estimation error is defined as $\tilde{\theta}_i = \theta_i^* - \hat{\theta}_i$. The basis function vector is $\phi_i = [\phi_{1,i}, \ldots, \phi_{M,i}]^T$, where $\phi_{m,i}$ is defined as follows:

$$\phi_{m,i}(z_i) = \frac{\prod_{k=1}^{j}\mu_{\Lambda_k^m}(z_{k,i})}{\sum_{s=1}^{M}\prod_{k=1}^{j}\mu_{\Lambda_k^s}(z_{k,i})} \tag{8}$$

where Λ_k^m is fuzzy set and $\mu_{\Lambda_k^m}(\cdot)$ is membership function.

Assumption 3 [40]: Both the minimum approximation error ϵ_i and the actual approximation error ν_i are bounded, and $\varpi_i = \epsilon_i - \nu_i$ is also bounded. That is, there are positive constants $\bar{\epsilon}_i$, $\bar{\nu}_i$, and $\bar{\varpi}_i$ satisfying $|\epsilon_i| \le \bar{\epsilon}_i$, $|\nu_i| \le \bar{\nu}_i$, and $|\varpi_i| \le \bar{\varpi}_i$.

III. Main Results

A. Extended State Fuzzy Observer Design

Due to uncertain internal dynamics and external disturbances, it is difficult to obtain complete and accurate vehicle motion information. To address this issue, an extended state fuzzy observer is designed for the ith vehicle, in which fuzzy logic systems approximate the unmodeled nonlinear dynamics and augmented states estimate the external disturbance, thereby enabling accurate reconstruction of the vehicle's position, velocity, acceleration, and disturbance

$$\begin{cases} \dot{\hat{x}}_{1,i} = \hat{x}_{2,i} + \hat{h}_{1,i}\left(\hat{\bar{x}}_{1,i}\right) + k_{1,i}\left(y_{2,i} - \hat{x}_{1,i}\right) \\ \dot{\hat{x}}_{2,i} = \hat{x}_{3,i} + \hat{h}_{2,i}\left(\hat{\bar{x}}_{2,i}\right) + k_{2,i}\left(y_{2,i} - \hat{x}_{1,i}\right) \\ \dot{\hat{x}}_{3,i} = \hat{x}_{4,i} + \frac{u_i}{m_i\tau_i} + \hat{h}_{3,i}\left(\hat{\bar{x}}_{3,i}\right) + k_{3,i}\left(y_{2,i} - \hat{x}_{1,i}\right) \\ \dot{\hat{x}}_{4,i} = \hat{x}_{5,i} + k_{4,i}\left(y_{2,i} - \hat{x}_{1,i}\right) \\ \dot{\hat{x}}_{5,i} = k_{5,i}\left(y_{2,i} - \hat{x}_{1,i}\right) \end{cases} \tag{9}$$

Matrix form is expressed as follows:

$$\dot{\hat{X}}_i = \Gamma_i\hat{X}_i + K_i y_{2,i} + \sum_{k=1}^{3}\mathcal{G}_k\hat{h}_{k,i}\left(\hat{\bar{x}}_{k,i}\right) + \mathcal{G}_3\frac{u_i}{m_i\tau_i} \tag{10}$$

where $\hat{x}_{k,i}$ $(k = 1, 2, 3)$ denote the estimates of $x_{k,i}$, $\hat{x}_{4,i}$ is the estimate of external disturbance D_i, and $\hat{x}_{5,i}$ is an auxiliary state to estimate $\dot{D}_i$, i.e., the dynamic variation of D_i. The coefficients $k_{k,i}$ are observer feedback gains, and $\hat{h}_{k,i}(\hat{\bar{x}}_{k,i}) = \hat{\theta}_{k,i}^T\phi_{k,i}(\hat{\bar{x}}_{k,i})$ represent the fuzzy approximations of $h_{k,i}(\hat{\bar{x}}_{k,i})$. In addition, $\mathcal{G}_1 = [1, 0, 0]^T$, $\mathcal{G}_2 = [0, 1, 0]^T$, $\mathcal{G}_3 = [0, 0, 1]^T$. $K_i = [k_{1,i},\ k_{2,i},\ k_{3,i},\ k_{4,i},\ k_{5,i}]^T$, $\Gamma_i = \begin{pmatrix} -k_{1,i} & 1 & 0 & 0 & 0 \\ -k_{2,i} & 0 & 1 & 0 & 0 \\ -k_{3,i} & 0 & 0 & 1 & 0 \\ -k_{4,i} & 0 & 0 & 0 & 1 \\ -k_{5,i} & 0 & 0 & 0 & 0 \end{pmatrix}$, $i = 1, 2, \ldots, n$.

Simultaneously, according to (1), (3), and (5), the extended matrix form of system (1) can be expressed as follows:

$$\dot{X}_i = \Gamma_i X_i + K_i y_{2,i} + \sum_{k=1}^{3} \mathcal{G}_k \left(\theta_{k,i}^{*T}\phi_{k,i}\left(\hat{\bar{x}}_{k,i}\right) + \epsilon_{k,i} + \Delta h_{k,i}\right) + \mathcal{G}_3\left(\frac{g_{0,i}u_i^c}{m_i\tau_i} + \frac{u_{di}}{m_i\tau_i}\right) \quad (11)$$

where $X_i = [x_{1,i},\ x_{2,i},\ x_{3,i},\ x_{4,i},\ x_{5,i}]^T$ with $x_{4,i} = D_i$. $\Delta h_{k,i} = h_{k,i}\left(\bar{x}_{k,i}\right) - h_{k,i}\left(\hat{\bar{x}}_{k,i}\right)$, and $h_{k,i}\left(\hat{\bar{x}}_{k,i}\right) = \theta_{k,i}^{*T}\phi_{k,i}\left(\hat{\bar{x}}_{k,i}\right) + \epsilon_{k,i}\ (k = 1, 2, 3)$.

Then, from (10) and (11), it follows that

$$\dot{\tilde{X}}_i = \Gamma_i\tilde{X}_i + \sum_{k=1}^{3}\mathcal{G}_k\left(\tilde{\theta}_{k,i}^T\phi_{k,i}\left(\hat{\bar{x}}_{k,i}\right) + \epsilon_{ki} + \Delta h_{k,i}\right) \quad (12)$$

where the state estimation error is defined as $\tilde{X}_i = [\tilde{x}_{1,i},\ \tilde{x}_{2,i},\ \tilde{x}_{3,i},\ \tilde{x}_{4,i},\ \tilde{x}_{5,i}]^T$.

For stability analysis of the extended state fuzzy observer (9), the following Lyapunov function candidate is constructed:

$$V_{0,i} = \frac{1}{2}\tilde{X}_i^T P_i \tilde{X}_i. \quad (13)$$

Substituting (12) into the time derivative of (13) yields

$$\begin{aligned}\dot{V}_{0i} &= \frac{1}{2}\dot{\tilde{X}}_i^T P_i\tilde{X}_i + \frac{1}{2}\tilde{X}_i^T P_i\dot{\tilde{X}} \\ &= \tilde{X}_i^T P_i\sum_{k=1}^{3}\mathcal{G}_k\left(\tilde{\theta}_{k,i}^T\phi_{k,i}\left(\hat{\bar{x}}_{k,i}\right) + \epsilon_{ki} + \Delta h_{ki}\right) \\ &\quad + \frac{1}{2}\tilde{X}_i^T\left(\Gamma_i^T P_i + P_i\Gamma_i\right)\tilde{X}_i. \quad (14)\end{aligned}$$

For a Hurwitz matrix Γ_i composed of coefficients $k_{k,i}$, and a positive definite matrix $Q_i = Q_i^T > 0$, there exists a positive definite matrix P_i satisfying $\Gamma_i^T P_i + P_i\Gamma_i = -Q_i$. Consequently, the following inequality holds:

$$\frac{1}{2}\tilde{X}_i^T\left(\Gamma_i^T P_i + P_i\Gamma_i\right)\tilde{X}_i \leq -\frac{\lambda_{\min}\left(Q_i\right)}{2}\|\ \tilde{X}_i\|^2. \quad (15)$$

By applying Young's inequality and using the property of the basis function $\phi_{k,i}^T\left(\hat{\bar{x}}_{k,i}\right)\phi_{k,i}\left(\hat{\bar{x}}_{k,i}\right) \leq 1$, we obtain

$$\tilde{X}_i^T P_i\sum_{k=1}^{3}\mathcal{G}_k\tilde{\theta}_{k,i}^T\phi_{k,i} \leq \frac{1}{2}\|\tilde{X}_i\|^2\|P_i\|^2 + \frac{1}{2}\sum_{k=1}^{3}\tilde{\theta}_{k,i}^T\tilde{\theta}_{k,i} \quad (16)$$

$$\tilde{X}_i^T P_i\sum_{k=1}^{3}\mathcal{G}_k\epsilon_{ki} \leq \frac{1}{2}\|\tilde{X}_i\|^2\|P_i\|^2 + \frac{1}{2}\sum_{k=1}^{3}\bar{\epsilon}_{k,i}^2 \quad (17)$$

$$\tilde{X}_i^T P_i\sum_{k=1}^{3}B_k\Delta h_{ki} \leq \frac{1}{2}\|\tilde{X}_i\|^2\|P_i\|^2 + \frac{1}{2}\sum_{k=1}^{3}r_{k,i}^2\|\tilde{X}_i\|^2. \quad (18)$$

Substituting inequalities (15)–(18) into (14) yields the following result:

$$\dot{V}_{0,i} \leq -\lambda_{0,i}\|\tilde{X}_i\|^2 + \frac{1}{2}\sum_{k=1}^{3}\tilde{\theta}_{k,i}^T\tilde{\theta}_{k,i} + D_{0,i} \quad (19)$$

where $\lambda_0 = \left((\lambda_{\min}(Q_i)/2) - (3/2)\|P_i\|^2 - (1/2)\sum_{k=1}^{3}r_{k,i}^2\right)\|\tilde{X}_i\|^2$, $D_{0,i} = (1/2)\sum_{k=1}^{3}\bar{\epsilon}_{k,i}^2$. Subsequently, the boundedness of $\tilde{\theta}_{k,i}$ will be proven in Section IV, where the stability of the observer will also be demonstrated.

B. Backstepping Controller Design

Within a backstepping framework, this section develops an extended observer-based adaptive fuzzy AFTC strategy for the longitudinal vehicle dynamics (1). Specifically, saturated Nussbaum techniques are employed to suppress actuator faults, and an adaptive compensation mechanism is designed to correct faulty inter-vehicle distance measurements.

Based on the CD policy and communication strategy illustrated in Fig. 1, the coordinate transformations for the ith vehicle are designed as:

$$e_{1,i} = y_{1,i} - d_i \quad (20)$$
$$e_{2,i} = \hat{x}_{2,i-1} - \hat{x}_{2,i} - \alpha_{1,i} \quad (21)$$
$$e_{3,i} = \hat{x}_{3,i-1} - \hat{x}_{3,i} - \alpha_{2,i} \quad (22)$$

where d_i denotes the required safe inter-vehicle spacing; $\alpha_{1,i}$ and $\alpha_{2,i}$ are the virtual control laws.

Step 1: Due to onboard ranging sensor faults, the true relative distance is unavailable for control. Therefore, the following transformation becomes necessary:

$$e_{1,i} = b_i y_{1,i}^f - d_i = \hat{b}_i y_{1,i}^f - d_i + \tilde{b}_i y_{1,i}^f \quad (23)$$
$$\hat{e}_{1,i} = \hat{b}_i y_{1,i}^f - d_i \quad (24)$$

where $b_i = 1/g_{1,i}$ is the compensation coefficient for the sensor effectiveness factor, $\hat{b}_i = b_i - \tilde{b}_i$ is the estimate of b_i with $\tilde{b}_i$ being the compensation error.

Differentiating $e_{1,i}$ yields

$$\dot{e}_{1,i} = x_{2,i-1} + h_{1,i-1} - \left(x_{2,i} + h_{1,i}\left(\hat{\bar{x}}_{1,i}\right) + \Delta h_{1,i}\right). \quad (25)$$

By replacing the true vehicular states with their observed counterparts and incorporating fuzzy approximation techniques (5) and (6), we obtain

$$\begin{aligned}\dot{e}_{1,i} &= \hat{x}_{2,i-1} + \tilde{x}_{2,i-1} + \hat{\theta}_{1,i-1}^T\phi_{1,i-1} + \nu_{1,i-1} \\ &\quad - \left(\hat{x}_{2,i} + \theta_{1,i}^{*T}\phi_{1,i} + \tilde{x}_{2,i} + \epsilon_{1,i} + \Delta h_{1,i}\right). \quad (26)\end{aligned}$$

Based on Lyapunov stability theory, a Lyapunov function incorporating a cubic term is selected to construct an adaptive compensation mechanism

$$V_{1,i} = V_{0,i} + \frac{1}{2}e_{1,i}^2 + \frac{1}{2\eta_{1,i}}\tilde{\theta}_{1,i}^T\tilde{\theta}_{1,i} + \frac{1}{3\gamma_i}\left|\tilde{b}_i\right|^3 \quad (27)$$

where η_{1i} and γ_{1i} are positive design constants. Substituting (26) into the time derivative of $V_{1,i}$, we obtain

$$\begin{aligned}\dot{V}_{1,i} &= \dot{V}_{0,i} + e_{1,i}\dot{e}_{1,i} - \frac{1}{\eta_{1,i}}\tilde{\theta}_{1,i}^T\dot{\hat{\theta}}_{1,i} - \frac{1}{\gamma_i}\tilde{b}_i^2\dot{\hat{b}}_i\,\mathrm{sign}\left(\tilde{b}_i\right) \\ &= \dot{V}_{0,i} + e_{1,i}\left(\alpha_{1,i} + \hat{\theta}_{1,i-1}^T\phi_{1,i-1} - \hat{\theta}_{1,i}^T\phi_{1,i}\right) \\ &\quad + e_{1,i}\left(e_{2,i} + \tilde{x}_{2,i-1} + \nu_{1,i-1} - \Delta h_{1,i} - \tilde{x}_{2,i} - \epsilon_{1,i}\right) \\ &\quad - e_{1,i}\tilde{\theta}_{1,i}^T\phi_{1,i} - \frac{1}{\eta_{1,i}}\tilde{\theta}_{1,i}^T\dot{\hat{\theta}}_{1,i} - \frac{1}{\gamma_i}\tilde{b}_i^2\dot{\hat{b}}_i\,\mathrm{sign}\left(\tilde{b}_i\right). \quad (28)\end{aligned}$$

Through the application of Young's inequality, the following inequalities can be obtained

$$e_{1,i}\left(e_{2,i}-\Delta h_{1,i}-\tilde{x}_{2,i}-\epsilon_{1,i}\right)\le 2e_{1,i}^2+\frac{1}{2}e_{2,i}^2+\frac{\bar{\epsilon}_{1,i}^2}{2}+\frac{1+r_{1,i}^2}{2}\left\|\tilde{X}_i\right\|^2 \tag{29}$$

$$e_{1,i}\left(\tilde{x}_{2,i-1}+\nu_{1,i-1}\right)\le e_{1,i}^2+\frac{1}{2}\left\|\tilde{X}_{i-1}\right\|^2+\frac{1}{2}\bar{\nu}_{1,i-1}^2 \tag{30}$$

$$-e_{1,i}\tilde{\theta}_{1,i}^T\phi_{1,i}\le -\hat{e}_{1,i}\tilde{\theta}_{1,i}^T\phi_{1,i}+\frac{1}{2}\left(\tilde{b}_i y_{1,i}^f\right)^2+\frac{1}{2}\tilde{\theta}_{1,i}^T\tilde{\theta}_{1,i}. \tag{31}$$

Subsequently, (28) is combined with the above inequalities and reformulated as follows:

$$\begin{aligned}\dot{V}_{1,i}\le{}&\dot{V}_{0,i}+e_{1,i}\left(\alpha_{1,i}+\hat{\theta}_{1,i-1}^T\phi_{1,i-1}-\hat{\theta}_{1,i}^T\phi_{1,i}+3e_{1,i}\right)\\&-\hat{e}_{1,i}\tilde{\theta}_{1,i}^T\phi_{1,i}+\frac{1}{2}\left(\tilde{b}_i y_i^f\right)^2+\frac{1}{2}\tilde{\theta}_{1,i}^T\tilde{\theta}_{1,i}+\frac{1}{2}\|\tilde{X}_{i-1}\|^2\\&-\frac{1}{\eta_{1,i}}\tilde{\theta}_{1,i}^T\dot{\hat{\theta}}_{1,i}-\frac{1}{\gamma_i}\tilde{b}_i^2\dot{\hat{b}}_i\,\mathrm{sign}\left(\tilde{b}_i\right)+\left(1+r_{1,i}^2\right)\frac{\|\tilde{X}_i\|^2}{2}\\&+\frac{1}{2}\bar{\epsilon}_{1,i}^2+\frac{1}{2}\bar{\nu}_{1,i-1}^2+\frac{1}{2}e_{2,i}^2.\end{aligned} \tag{32}$$

The virtual control law and adaptive law for the ith vehicle are then designed as follows:

$$\alpha_{1,i}=-c_{0,i}\hat{e}_{1,i}-\hat{\theta}_{1,i-1}^T\phi_{1,i-1}+\hat{\theta}_{1,i}^T\phi_{1,i} \tag{33}$$

$$\dot{\hat{\theta}}_{1,i}=-\mu_{1,i}\hat{\theta}_{1,i}-\eta_{1,i}\hat{e}_{1,i}\phi_{1,i} \tag{34}$$

where $c_{0,i}>0$ and $\mu_{1,i}>0$ are user-defined parameters.

Then, (32) is reformulated as follows:

$$\begin{aligned}\dot{V}_{1,i}\le{}&\dot{V}_{0,i}+e_{1,i}\left(-c_{0,i}\hat{e}_{1,i}+3e_{1,i}\right)+\frac{\mu_{1,i}}{\eta_{1,i}}\tilde{\theta}_{1,i}^T\hat{\theta}_{1,i}+\frac{1}{2}\tilde{\theta}_{1,i}^T\tilde{\theta}_{1,i}\\&+\frac{1}{2}\left(\tilde{b}_i y_i^f\right)^2-\frac{1}{\gamma_i}\tilde{b}_i^2\dot{\hat{b}}_i\,\mathrm{sign}\left(\tilde{b}_i\right)+\left(1+r_{1,i}^2\right)\frac{\|\tilde{X}_i\|^2}{2}\\&+\frac{1}{2}\|\tilde{X}_{i-1}\|^2+\frac{1}{2}\bar{\nu}_{1,i-1}^2+\frac{1}{2}\bar{\epsilon}_{1,i}^2+\frac{1}{2}e_{2,i}^2\end{aligned} \tag{35}$$

with the following inequality exists:

$$\begin{aligned}-c_{0,i}e_{1,i}\hat{e}_{1,i}&=-c_{0,i}e_{1,i}^2+c_{0,i}e_{1,i}\tilde{b}_i y_{1,i}^f\\&=-c_{0,i}e_{1,i}^2+c_{0,i}\left(\tilde{b}_i y_{1,i}^f\right)^2+c_{0,i}\hat{e}_{1,i}\tilde{b}_i y_{1,i}^f\\&\le -c_{0,i}e_{1,i}^2+\frac{1}{2\delta_i}\\&\quad+\left(c_{0,i}+\frac{\delta_i}{2}\left(c_{0,i}\hat{e}_{1,i}\right)^2\right)\left(\tilde{b}_i y_{1,i}^f\right)^2.\end{aligned} \tag{36}$$

Consequently, (35) can be rewritten as follows:

$$\begin{aligned}\dot{V}_{1,i}\le{}&\dot{V}_{0,i}+e_{1,i}\left(-c_{0,i}e_{1,i}+3e_{1,i}\right)+\frac{\mu_{1,i}}{\eta_{1,i}}\tilde{\theta}_{1,i}^T\hat{\theta}_{1,i}\\&+\frac{1}{2}\tilde{\theta}_{1,i}^T\tilde{\theta}_{1,i}+\left(c_{0,i}+\frac{\delta_i}{2}\left(c_{0,i}\hat{e}_{1,i}\right)^2+\frac{1}{2}\right)\left(\tilde{b}_i y_{1,i}^f\right)^2\\&-\frac{1}{\gamma_i}\tilde{b}_i^2\dot{\hat{b}}_i\,\mathrm{sign}\left(\tilde{b}_i\right)+\left(1+r_{1,i}^2\right)\frac{\|X_i\|^2}{2}\\&+\frac{1}{2}\|X_{i-1}\|^2+\frac{1}{2\delta_i}+\frac{1}{2}\bar{\nu}_{1,i-1}^2+\frac{1}{2}\bar{\epsilon}_{1,i}^2+\frac{1}{2}e_{2,i}^2.\end{aligned} \tag{37}$$

Remark 5: Due to onboard ranging sensor faults, the actual tracking error $e_{1,i}$ is inaccessible. Therefore, the inequality scaling in (36) is necessary to prevent $e_{1,i}$ from being used in either the controller or the adaptation law, and the virtual controller (33) should be deliberately constructed using $\hat{e}_{1,i}$.

Subsequently, the adaptive law for the compensation coefficient is designed as follows:

$$\begin{aligned}\dot{\hat{b}}_i&=\mathrm{Proj}_{\left[1,\,\frac{1}{g_{1,i}}\right]}\{B_i\}\\&=\begin{cases}0, & \hat{b}_i=\frac{1}{g_{1,i}}\text{ and }B_i\ge 0,\hat{b}_i=1\text{ and }B_i\le 0\\ B_i, & \text{otherwise}\end{cases}\end{aligned} \tag{38}$$

with

$$B_i=\begin{cases}\Xi_i, & \Xi_i\ge 0\\ 0, & \Xi_i<0\end{cases}$$

$$\Xi_i=-\sigma_i\hat{b}_i+\gamma_i\left[\delta_i(c_{0,i}\hat{e}_{1,i})^2+c_{0,i}+\tfrac{1}{2}\right](y_{1,i}^f)^2$$

where $\mathrm{Proj}[\cdot]$ denotes the projection operator that constrains the estimate $\hat{b}_i$ within the interval $[1,\ 1/g_{1,i}]$ $\sigma_i>0$ and $\delta_i>0$ are design parameters.

Remark 6: It follows from (38) that the compensation term is explicitly scaled by $c_{0,i}$. Consequently, the accuracy of the compensation becomes coupled to the choice of the controller gain: if $c_{0,i}$ is chosen too small, the compensation reacts sluggishly, whereas an excessively large $c_{0,i}$ may lead to overcompensation. To mitigate this coupling and enhance the sensitivity of the mechanism to faulty distance information, δ_i is selected sufficiently large and γ_i sufficiently small. In this way, the mechanism can exploit the rapid variation of $\hat{e}_{1,i}$ at the onset of faults and ensures effective compensation with adaptability to different fault conditions.

Subsequently, combining with (19) and (38), (37) is reformulated as follows:

$$\begin{aligned}\dot{V}_{1,i}\le{}&\Upsilon_i-c_{1,i}e_{1,i}^2+\frac{\mu_{1,i}}{\eta_{1,i}}\tilde{\theta}_{1,i}^T\hat{\theta}_{1,i}+\frac{\sigma_{1,i}}{\gamma_{1,i}}\tilde{b}_i^2\hat{b}_i\,\mathrm{sign}\left(\tilde{b}_i\right)\\&+D_{1,i}+\frac{1}{2}e_{2,i}^2\end{aligned} \tag{39}$$

where $\Upsilon_i=-\lambda_{1,i}\|\tilde{X}_i\|^2+(1/2)\|\tilde{X}_{i-1}\|^2+(1/2)\sum_{k=1}^3\tilde{\theta}_{k,i}^T\tilde{\theta}_{k,i}+(1/2)\tilde{\theta}_{1,i}^T\tilde{\theta}_{1,i}$, $c_{1,i}=c_{0,i}-3$, $\lambda_{1,i}=\lambda_{0,i}-(1/2)\left(1+r_{1,i}^2\right)\|\tilde{X}_i\|^2$, $D_{1,i}=D_{0,i}+(1/2)\bar{\epsilon}_{1,i}^2+(1/2\delta_i)+(1/2)\bar{\nu}_{1,i-1}^2$.

Step 2: Differentiating $e_{2,i}=\hat{x}_{2,i-1}-\hat{x}_{2,i}-\alpha_{1,i}$ and combining with (9) yields

$$\begin{aligned}\dot{e}_{2,i}={}&\hat{x}_{3,i-1}+\hat{h}_{2,i-1}\left(\hat{\bar{x}}_{2,i-1}\right)+k_{2,i-1}\left(y_{2,i-1}-\hat{x}_{1,i-1}\right)\\&-\left(\hat{x}_{3,i}+\hat{h}_{2,i}\left(\hat{\bar{x}}_{2,i}\right)+k_{2,i}\left(y_{2,i}-\hat{x}_{1,i}\right)\right)-\dot{\alpha}_{1,i}.\end{aligned} \tag{40}$$

The Lyapunov function candidate is chosen as follows:

$$V_{2,i}=V_{1,i}+\frac{1}{2}e_{2,i}^2+\frac{1}{2\eta_{2,i}}\tilde{\theta}_{2,i}^T\tilde{\theta}_{2,i}+\frac{1}{2l_{2,i}}\tilde{\Delta}_{2,i}^2 \tag{41}$$

in which $\eta_{2,i}$ and $l_{2,i}$ denote positive design constants. Substituting (41) into $\dot{V}_{2,i}$, it follows that

$$\begin{aligned}\dot{V}_{2,i} = \dot{V}_{1,i} + e_{2,i}\left(\alpha_{2,i} + \hat{\theta}_{2,i-1}^T\phi_{2,i-1} - k_{2,i}\left(y_{2,i} - \hat{x}_{1,i}\right)\right)\\ - e_{2,i}\hat{\theta}_{2,i}\phi_{2,i} + e_{2,i}\left(e_{3,i} - \varpi_{2,i}\right) + e_{2,i}H_{2,i}\\ + \tilde{\theta}_{2,i}^T\left(-e_{2,i}\phi_{2,i} - \frac{1}{\eta_{2,i}}\dot{\hat{\theta}}_{2,i}\right) - \frac{1}{l_{2,i}}\tilde{\Delta}_{2,i}\dot{\hat{\Delta}}_{2,i} \qquad (42)\end{aligned}$$

where $H_{2,i} = k_{2,i-1}\left(y_{2,i-1} - \hat{x}_{1,i-1}\right) - \dot{\alpha}_{1,i}$

Substituting (39) into (42), it yields

$$\begin{aligned}\dot{V}_{2,i} \le \Upsilon_i + \frac{\mu_{1,i}}{\eta_{1,i}}\tilde{\theta}_{1,i}^T\hat{\theta}_{1,i} + \frac{\sigma_{1,i}}{\gamma_{1,i}}\tilde{b}_i^2\hat{b}_i\,\mathrm{sign}\left(\tilde{b}_i\right) + D_{1,i} + \frac{1}{2}e_{2,i}^2\\ + e_{2,i}\left(\alpha_{2,i} + \hat{\theta}_{2,i-1}^T\phi_{2,i-1} - k_{2,i}\left(y_{2,i} - \hat{x}_{1,i}\right)\right)\\ - c_{1,i}e_{1,i}^2 - e_{2,i}\hat{\theta}_{2,i}\phi_{2,i} + e_{2,i}\left(e_{3,i} - \varpi_{2,i}\right) + e_{2,i}H_{2,i}\\ + \tilde{\theta}_{2,i}^T\left(-e_{2,i}\phi_{2,i} - \frac{1}{\eta_{2,i}}\dot{\hat{\theta}}_{2,i}\right) - \frac{1}{l_{2,i}}\dot{\hat{\Delta}}_{2,i}. \qquad (43)\end{aligned}$$

According to Lemma 2, $H_{2,i}$ is capable of being approximated through a fuzzy logic system as: $H_{2,i} = W_{2,i}^{*T}\kappa_{2,i} + o_{2,i}$, where $o_{2,i}$ is the approximation error with $|o_{2,i}| \le \bar{o}_{2,i}$.

Therefore, the following inequalities holds

$$e_{2,i}\left(e_{3,i} - \varpi_{2,i}\right) \le e_{2,i}^2 + \frac{1}{2}\bar{\varpi}_{2,i}^2 + \frac{1}{2}e_{3,i}^2 \qquad (44)$$

$$e_{2,i}H_{2,i} \le \frac{\xi_{2,i}e_{2,i}^2\Delta}{2} + \frac{1}{2}e_{2,i}^2 + \frac{1}{2}\bar{o}_{2,i}^2 + \frac{1}{2\xi_{2,i}} \qquad (45)$$

where $\Delta = W_{2,i}^{*T}W_{2,i}^*$ and $\xi_{2,i} > 0$.

Therefore, we design the virtual controller $\alpha_{2,i}$, adaptive laws of $\theta_{2,i}$ and $\Delta_{2,i}$ as follows:

$$\begin{aligned}\alpha_{2,i} = -c_{2,i}e_{2,i} - 2e_{2,i} - \hat{\theta}_{2,i-1}^T\phi_{2,i-1} + \hat{\theta}_{2,i}^T\phi_{2,i}\\ + k_{2,i}\left(y_{2,i} - \hat{x}_{1,i}\right) - \frac{\xi_{2,i}}{2}e_{2,i}\hat{\Delta}_{2,i} \qquad (46)\end{aligned}$$

$$\dot{\hat{\theta}}_{2,i} = -\mu_{2,i}\hat{\theta}_{2,i} - \eta_{2,i}e_{2,i}\phi_{2,i} \qquad (47)$$

$$\dot{\hat{\Delta}}_{2,i} = -\pi_{2,i}\hat{\Delta}_{2,i} + \frac{1}{2}\xi_{2,i}l_{2,i}e_{2,i}^2 \qquad (48)$$

where $c_{2,i}$, $\mu_{2,i}$, and $\pi_{2,i}$ are positive user-defined parameters.

By substituting (44)–(48) into the time derivative of $V_{2,i}$, the following result is obtained

$$\begin{aligned}\dot{V}_{2,i} \le \Upsilon_i - \sum_{k=1}^{2}c_{k,i}e_{k,i}^2 + \sum_{k=1}^{2}\frac{\mu_{k,i}}{\eta_{k,i}}\tilde{\theta}_{k,i}^T\hat{\theta}_{k,i} + \frac{\sigma_i}{\gamma_i}\tilde{b}_i^2\hat{b}_i\,\mathrm{sign}\left(\tilde{b}_i\right)\\ + \frac{\pi_{2,i}}{l_{2,i}}\tilde{\Delta}_{2,i}\hat{\Delta}_{2,i} + D_{2,i} + \frac{1}{2}e_{3,i}^2 \qquad (49)\end{aligned}$$

where $D_{2,i} = D_{1,i} + (1/2)\bar{\varpi}_{2,i}^2 + (1/2)\bar{o}_{2,i}^2 + (1/2\xi_{2,i})$.

Step 3: Calculate the derivative of $e_{3,i}$:

$$\begin{aligned}\dot{e}_{3,i} = \frac{g_{0,i-1}u_{i-1}^c}{m_{i-1}\tau_{i-1}} + \hat{h}_{3,i-1}\left(\hat{\bar{x}}_{3,i-1}\right) - \dot{\alpha}_{2,i}\\ - \frac{g_{0,i}u_i^c}{m_i\tau_i} - \hat{h}_{3,i}\left(\hat{\bar{x}}_{3,i}\right) - k_{3,i}\left(y_{2,i} - \hat{x}_{1,i}\right)\\ + k_{3,i-1}\left(y_{2,i-1} - \hat{x}_{1,i-1}\right). \qquad (50)\end{aligned}$$

Proceed to construct the Lyapunov function following the structure established in Step 2:

$$V_{3,i} = V_{2,i} + \frac{1}{2}e_{3,i}^2 + \frac{1}{2\eta_{3,i}}\tilde{\theta}_{3,i}^T\tilde{\theta}_{3,i} + \frac{1}{2l_{3,i}}\tilde{\Delta}_{3,i}^2 + \frac{1}{2\beta_i}\tilde{g}_{0,i-1}^2 \qquad (51)$$

where $\eta_{3,i}$, $l_{3,i}$, and β_i are positive design constants. $\hat{g}_{0,i-1}$ denotes the estimate of $g_{0,i-1}$ and $\tilde{g}_{0,i-1} = g_{0,i-1} - \hat{g}_{0,i-1}$.

Integrating (50), the derivative of (51) emerges as follows:

$$\begin{aligned}\dot{V}_{3,i} = \dot{V}_{2i} + e_{3i}\Big(-\frac{g_{0,i}u_i^c}{m_i\tau_i} + \hat{\theta}_{3,i-1}^T\phi_{3,i-1} - \hat{\theta}_{3,i}^T\phi_{3,i} - \hat{D}_i\\ - k_{3,i}\left(y_{2,i} - \hat{x}_{1,i}\right)\Big) + e_{3i}g_{0,i-1}N_{i-1}\psi_{i-1} - e_{3i}\varpi_{3,i}\\ + \tilde{\theta}_{3,i}^T\left(-e_{3i}\phi_{3,i} - \frac{1}{\eta_{3,i}}\dot{\hat{\theta}}_{3,i}\right) + e_{3,i}H_{3,i}\\ - \frac{1}{l_{3,i}}\tilde{\Delta}_{2,i}\dot{\hat{\Delta}}_{3,i} - \frac{1}{\beta_i}\tilde{g}_{0,i-1}\dot{\hat{g}}_{0,i-1} \qquad (52)\end{aligned}$$

where $H_{3,i} = (u_{d,i-1}/m_{i-1}\tau_{i-1}) - (u_{d,i}/m_i\tau_i) + \hat{D}_{i-1} - k_{3,i-1}\left(y_{2,i-1} - \hat{x}_{1,i-1}\right) - \dot{\alpha}_{2,i}$. $H_{3,i}$ is modeled by a fuzzy logic system as: $H_{3,i} = W_{3,i}^{*T}\kappa_{3,i} + o_{3,i}$, where $o_{3,i}$ is the approximation error with $|o_{3,i}| \le \bar{o}_{3,i}$.

The terms involved in (52) are subject to the subsequent inequalities

$$-e_{3,i}\varpi_{3,i} \le \frac{1}{2}e_{3,i}^2 + \frac{1}{2}\bar{\varpi}_{3,i}^2 \qquad (53)$$

$$e_{3,i}H_{3,i} \le \frac{\xi_{3,i}e_{3,i}^2\Delta_{3,i}}{2} + \frac{1}{2}e_{3,i}^2 + \frac{1}{2}\bar{o}_{3,i}^2 + \frac{1}{2\xi_{3,i}}. \qquad (54)$$

Based on prior analysis, the actual controller and adaptive laws are then specified as follows:

$$u_i^c = m_i\tau_iN_i\Psi_i \qquad (55)$$

$$\begin{aligned}\Psi_i = -c_{3,i}e_{3,i} - \frac{3}{2}e_{3,i} - \frac{1}{2}e_{3,i}\xi_{3,i}\hat{\Delta}_{3,i} - \hat{g}_{0,i-1}N_{i-1}\Psi_{i-1}\\ - \hat{\theta}_{3,i-1}^T\phi_{3,i-1} + k_{3,i}\left(y_{2,i} - \hat{x}_{1,i}\right) + \hat{\theta}_{3,i}^T\phi_{3,i} + \hat{D}_i \qquad (56)\end{aligned}$$

$$\dot{\hat{\theta}}_{3,i} = -\mu_{3,i}\hat{\theta}_{3,i} - \eta_{3,i}e_{3,i}\phi_{3,i} \qquad (57)$$

$$\dot{\hat{\Delta}}_{3,i} = -\pi_{3,i}\hat{\Delta}_{3,i} + \frac{1}{2}\xi_{3,i}l_{3,i}e_{3,i}^2 \qquad (58)$$

$$\dot{\hat{g}}_{0,i-1} = -\omega_i\hat{g}_{0,i-1} + \beta_ie_{3,i}N_{i-1}\Psi_{i-1} \qquad (59)$$

where $c_{3,i}$, $\mu_{3,i}$, $\pi_{3,i}$, ω_i are positive user-defined parameters.

Substituting (53)–(59) into the derivative of $V_{3,i}$, the following result is obtained:

$$\begin{aligned}\dot{V}_{3,i} \le \Upsilon_i - \sum_{k=1}^{3}c_{k,i}e_{k,i}^2 + \left(g_{0,i}N_i + 1\right)\dot{\chi}_i + D_{3,i}\\ + \sum_{k=1}^{3}\frac{\mu_{k,i}}{\eta_{k,i}}\tilde{\theta}_{k,i}^T\hat{\theta}_{k,i} + \sum_{k=2}^{3}\frac{\pi_{k,i}}{l_{k,i}}\tilde{\Delta}_{k,i}\hat{\Delta}_{k,i}\\ + \frac{\omega_i}{\beta_i}\tilde{g}_{0,i-1}\hat{g}_{0,i-1} + \frac{\sigma_i}{\gamma_i}\tilde{b}_i^2\hat{b}_i\,\mathrm{sign}\left(\tilde{b}_i\right) \qquad (60)\end{aligned}$$

where $D_{3,i} = D_{2,i} + (1/2)\bar{\varpi}_{3,i}^2 + (1/2)\bar{o}_{3,i}^2 + (1/2\xi_{3,i})$, $\dot{\chi}_i = -e_{3,i}\Psi_i$.

IV. Stabilization Analysis

Theorem 1: Under Assumptions 1–3, for vehicle dynamics (1) subject to actuator faults (3) and ranging sensor faults (4), by employing the virtual controllers (33) and (46), the actual controller (55), the adaptive laws (34), (47), (48), (57), (58), and (59), the extended state fuzzy observer (9), and the compensation mechanism (38), the vehicular platoon achieves internal stability and string stability.

Proof: From (38), it is seen that $\hat{b}_i \geq 0$, which implies $\hat{b}_i(t) > 0$ for all $t > 0$ only if $\hat{b}_i(0) \geq 0$. In fact, it is always reasonable to choose $\hat{b}_i(0) \geq 0$ in practice, since $\hat{b}_i$ is an estimate of b_i. Noting that $\dot{\hat{b}}_i \geq 0$ and $\tilde{b}_i = b_i - \hat{b}_i$, with $\lim_{t\to\infty}\hat{b}_i(t) = b_i$, it follows that $\hat{b}_i(t) < b_i$, i.e., $\tilde{b}_i(t) > 0$ for all $t \geq 0$ only if $\hat{b}_i(0) < b_i$. Thus, the following inequalities holds:

$$\tilde{\theta}_{k,i}^T\hat{\theta}_{k,i} \leq -\frac{1}{2}\tilde{\theta}_{k,i}^T\tilde{\theta}_{k,i} + \frac{1}{2}\theta_{k,i}^{*T}\theta_{k,i}^* \tag{61}$$

$$\tilde{b}_i^2\hat{b}_i \operatorname{sign}\left(\tilde{b}_i\right) \leq -\frac{1}{3}\left|\tilde{b}_i^3\right| + \frac{1}{3}b_i^3 \tag{62}$$

$$\tilde{\Delta}_{k,i}\hat{\Delta}_{k,i} \leq -\frac{1}{2}\tilde{\Delta}_{k,i}^2 + \frac{1}{2}\Delta_{k,i}^2 \tag{63}$$

$$\tilde{g}_{0,i-1}\hat{g}_{0,i-1} \leq -\frac{1}{2}\tilde{g}_{0,i-1}^2 + \frac{1}{2}g_{0,i-1}^2. \tag{64}$$

Let $V_i = V_{3,i}$. Hence, from (60), it is obtained that

$$\begin{aligned}\dot{V}_i \leq& -\lambda_{1,i}\|\tilde{X}_i\|^2 + \frac{1}{2}\|\tilde{X}_{i-1}\|^2 - \sum_{k=1}^{3}c_{k,i}e_{k,i}^2\\&-\sum_{k=1}^{3}\left(\frac{\mu_{k,i}}{2\eta_{k,i}} - 1\right)\tilde{\theta}_{k,i}^T\tilde{\theta}_{k,i} - \frac{1}{2}\sum_{k=2}^{3}\frac{\pi_{k,i}}{l_{k,i}}\tilde{\Delta}_{k,i}^2\\&+\left(g_iN_i + 1\right)\dot{\chi}_i - \frac{\omega_i}{2\beta_i}\tilde{g}_{0,i-1}^2 - \frac{\sigma_i}{3\gamma_i}\left|\tilde{b}_i^3\right| + D_i^*\end{aligned} \tag{65}$$

where $D_i^* = D_{3,i} + (\sigma_i/3\gamma_i)b_i^3 + (\omega_i/2\beta_i)g_{0,i-1}^2 + \sum_{k=1}^{3}(\mu_{k,i}/2\eta_{k,i})\theta_{k,i}^{*T}\theta_{k,i}^* + \sum_{k=2}^{3}(\pi_{k,i}/2l_{k,i})\Delta_{k,i}^2$.

Next, the whole Lyapunov function for the vehicular platoon is formulated as $V = \sum_{i=1}^{n} V_i$. After calculation, we obtain

$$\begin{aligned}V =& \sum_{i=1}^{n}\left(\frac{1}{2}\tilde{X}_i^TP_i\tilde{X}_i + \frac{1}{3\gamma_i}\left|\tilde{b}_i\right|^3 + \frac{1}{2\beta_i}\tilde{g}_{i-1}^2\right)\\&+\sum_{i=1}^{n}\sum_{k=1}^{3}\frac{1}{2}e_{k,i}^2 + \sum_{i=1}^{n}\sum_{k=1}^{3}\frac{1}{2}\tilde{\theta}_{k,i}^T\tilde{\theta}_{k,i} + \sum_{i=1}^{n}\sum_{k=2}^{3}\frac{1}{2}\tilde{\Delta}_{k,i}^2.\end{aligned} \tag{66}$$

This readily yields:

$$\begin{aligned}\dot{V} \leq& \sum_{i=1}^{n}\Bigg(-\overline{\lambda}_{1,i}\|\tilde{X}_i\|^2 - \sum_{k=1}^{3}c_{k,i}e_{k,i}^2 + \left(g_iN_i + 1\right)\dot{\chi}_i\\&-\sum_{k=1}^{3}\left(\frac{\mu_{k,i}}{2\eta_{k,i}} - 1\right)\tilde{\theta}_{k,i}^T\tilde{\theta}_{k,i} - \frac{1}{2}\sum_{k=2}^{3}\frac{\pi_{k,i}}{l_{k,i}}\tilde{\Delta}_{k,i}^2\\&-\frac{\omega_i}{2\beta_i}\tilde{g}_{0,i-1}^2 - \frac{\sigma_i}{3\gamma_i}\left|\tilde{b}_i^3\right| + D_i^*\Bigg)\\\leq& -CV + \sum_{i=1}^{n}\left(g_iN_i + 1\right)\dot{\chi}_i + D\end{aligned} \tag{67}$$

where $C = \min\{(2\overline{\lambda}_{1,i}/\lambda_{\max}(P_i)),\ 2c_{k,i},\ \mu_{k,i} - 2\eta_{k,i},\ \sigma_i,\ \pi_{k,i},\ \omega_i\}$, $D = \sum_{i=1}^{n}D_i^*$, $\overline{\lambda}_{1,i} = \lambda_{1,i} - (1/2)$.

According to Lemma 1, $V(t)$, χ_i and $\sum_{i=1}^{n}(g_iN_i + 1)\dot{\chi}_i$ are bounded. Thus, there exists a positive constant $\aleph$ such that $\left|\sum_{i=1}^{n}(g_iN_i + 1)\dot{\chi}_i\right| \leq \aleph, \quad \forall t \geq 0$. Then, inequality (IV) becomes

$$\dot{V} \leq -CV + \overline{D} \tag{68}$$

where $\overline{D} = \aleph + D$. Therefore, the errors satisfy $|e_{k,i}| \leq \sqrt{2V(0)e^{-Ct} + 2\overline{D}/C}$, which implies that $\max_i|e_{1,i}|$ converges to a compact set centered at the origin with radius $\sqrt{2\overline{D}/C}$. According to Definition 1, the string stability of the vehicular platoon is thus guaranteed.

Remark 7: From (68), the tracking errors converge exponentially in the sense that $\limsup_{t\to\infty}\max_i|e_{1,i}| \leq \sqrt{2\overline{D}/C}$. By appropriately tuning the design parameters involved in C and $\overline{D}$, this ultimate bound can be made arbitrarily small, so that the spacing error remains close to zero. In particular, increasing the feedback gains $c_{k,i}$ and decreasing the adaptation gains $\mu_{k,i}$, $\pi_{k,i}$, σ_i, and ω_i can enlarge the decay rate C, whereas decreasing the constants contributing to $\overline{D}$ improves the steady-state tracking accuracy. It should also be noted that choosing σ_i too small may cause overcompensation, thereby degrading tracking performance. Therefore, these parameters should be selected carefully in practical implementations.

At the steady state, all vehicles travel at the same velocity $x_{2,0}$ while maintaining the constant spacing $l + d$. Therefore, the traffic density and traffic flow rate are given by

$$\rho = \frac{1}{l+d}, \qquad f = \frac{x_{2,0}}{l+d} = x_{2,0}\rho. \tag{69}$$

Since $x_{2,0} > 0$, it follows that

$$\frac{\partial f}{\partial \rho} = x_{2,0} > 0. \tag{70}$$

Hence, according to Definition 3, traffic flow stability is guaranteed. Therefore, the proposed AFTC scheme achieves internal stability, string stability and traffic flow stability simultaneously for the vehicular platoon.

V. Simulation Example

This section presents simulations conducted in Simulink (MATLAB R2024a) to verify the effectiveness of the proposed adaptive fuzzy AFTC, where the ode23t solver with the Refine output option is employed to improve efficiency. The vehicular platoon operates under the PF topology illustrated in Fig. 1. The leader vehicle dynamics are described by (2), while the follower vehicles, subject to actuator faults (3) and ranging sensor faults (4), are modeled as follows:

$$\begin{cases}\dot{x}_{1,i} = x_{2,i} + x_{1,i}e^{-x_{1,i}}\\\dot{x}_{2,i} = x_{3,i} + x_{2,i}e^{-x_{1,i}}\\\dot{x}_{3,i} = \frac{u_i}{m_i\tau_i} - \frac{k_{d,i}}{m_i\tau_i}(x_{2,i}^2 + 2\tau_ix_{2,i}x_{3,i})\\\qquad - \frac{x_{3,i}}{\tau_i} + D_i\end{cases}$$

TABLE I
DESIGN PARAMETERS AND INITIAL VALUES

Parameters	Value
Control parameters	$c_{0,i}=1,\ c_{1,1}=2,\ c_{1,2}=15,\ c_{1,3}=5$ $c_{1,4}=20,\ c_{1,5}=2,\ c_{1,6}=2,\ c_{1,7}=20$ $c_{2,1}=4,\ c_{2,2}=75,\ c_{2,3}=15,\ c_{2,4}=60$ $c_{2,5}=15,\ c_{2,6}=100,\ c_{2,7}=100$
Gain coefficients	$k_{1,1}=6,\ k_{2,i}=8,\ k_{3,i}=10$ $k_{4,i}=20,\ k_{5,i}=10$
Adaptive parameters 1	$\eta_{k,i}=1;\ \mu_{k,i}=1;\ \beta_i=1;\ \omega_i=1$ $l_{k,i}=1,\ \xi_{k,i}=1;\ \pi_{k,i}=1$
Adaptive parameters 2	$\sigma_1=1,\ \gamma_1=0.1;\ \delta_2=165$ $\sigma_3=1,\ \gamma_3=0.01;\ \delta_3=6000$
Initial values 1	$x_{1,0}(0)=70,\ x_{1,1}(0)=60,\ x_{1,2}(0)=50$ $x_{1,3}(0)=40,\ x_{1,4}(0)=30,\ x_{1,5}(0)=20$ $x_{1,6}(0)=10,\ x_{1,7}(0)=0$
Initial values 2	$x_{k,i}(0)=0,\ x_{k,i}(0)=1$ $\hat{b}_i(0)=1$

TABLE II
FAULT INFORMATION

Vehicle Index	Fault Information	Time When Fault Occurs
1	$g_{0,1}=0.5$	$t>40$ s
	$g_{1,1}=0.8$	$t>0$ s
	$u_{d,1}=20sin(t)$	$t>0$ s
2	$g_{0,2}=0.8$	$t>60$ s
	$u_{d,2}=40$	$t>0$ s
3	$g_{0,3}=0.6$	$t>40$ s
	$g_{1,3}=0.5$	$t>0$ s
	$u_{d,3}=10sin(t)$	$t>0$ s
4	$g_{0,4}=0.2$	$t>0$ s
5	$g_{0,5}=0.2$	$t>40$ s
6	$g_{0,4}=0.7$	$t>40$ s
7	$g_{0,4}=0.1$	$t>0$ s

Set the membership function as: $\mu_{\Lambda_k^m}\left(\hat{x}_{k,i}\right)=\exp\left(-\left(\left(\hat{x}_{k,i}-2+m\right)^2/2\right)\right), m=1,2,3.$

The corresponding normalized fuzzy basis functions are selected as: $\phi_{m,i}\left(\hat{\bar{x}}_{1,i}\right)=\left(\mu_{\Lambda_1^m}\left(\hat{x}_{1,i}\right)/\sum_{s=1}^{3}\mu_{\Lambda_1^s}\left(\hat{x}_{1,i}\right)\right)$, $\phi_{m,i}\left(\hat{\bar{x}}_{2,i}\right)=\left(\prod_{k=1}^{2}\mu_{\Lambda_k^m}\left(\hat{x}_{k,i}\right)/\sum_{s=1}^{3}\prod_{k=1}^{2}\mu_{\Lambda_k^s}\left(\hat{x}_{k,i}\right)\right)$, $\phi_{m,i}\left(\hat{\bar{x}}_{3,i}\right)=\left(\prod_{k=1}^{3}\mu_{\Lambda_k^m}\left(\hat{x}_{k,i}\right)/\sum_{s=1}^{3}\prod_{k=1}^{3}\mu_{\Lambda_k^s}\left(\hat{x}_{k,i}\right)\right)$.

The system parameters are chosen as $m_i=1000$kg, $\tau_i=0.25$s, $k_{d,i}=0.4$, $l_{i-1}=3$m, and $d_i=7$m. Table I presents the design parameters and initial conditions. Furthermore, to evaluate the performance of the vehicular platoon under adverse conditions, different fault scenarios are considered, as listed in Table II. In addition, to verify the robustness of the proposed AFTC scheme, the external disturbances are set as $D_1=2\sin(0.1t)$, $D_2=5\sin(0.1t)$, $D_3=10\sin(0.1t)$, and $D_4=D_5=D_6=D_7=3\sin(0.1t)$. The simulation results are shown in Figs. 2–10.

As shown in Fig. 2, all follower vehicles maintain the prescribed safe spacing under the CD policy in the presence of hybrid fault scenarios. Furthermore, Figs. 3 and 4(a) show that each follower vehicle accurately tracks the predecessor's velocity and acceleration, which confirms that the proposed adaptive fuzzy AFTC scheme can effectively handle dual-fault while preserving the internal stability of the vehicular platoon.

In addition, the proposed adaptive fuzzy AFTC scheme is further evaluated by replacing the saturated Nussbaum function with the conventional one in [40], chosen as $N_i=\cos\left(X_i\right)X_i^2$, while keeping all other conditions unchanged. The corresponding acceleration responses are shown in Fig. 4(b). As can be seen, the conventional Nussbaum function causes significant

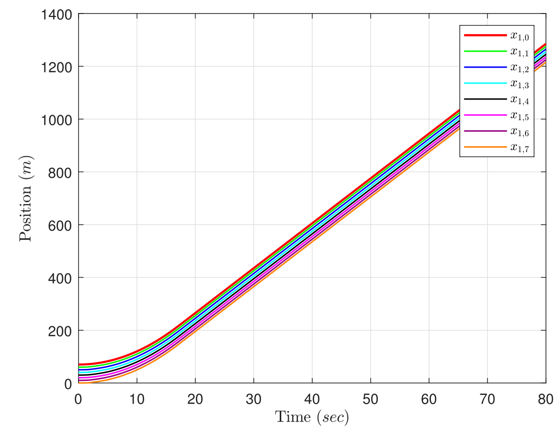

Fig. 2. Results of vehicular positions $x_{1,i}, i\in\{0,1,\ldots,7\}$ on leader and follower vehicles.

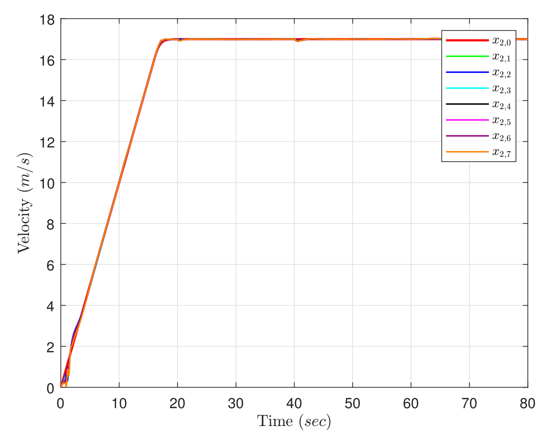

Fig. 3. Results of vehicular velocities $x_{2,i}, i\in\{0,1,\ldots,7\}$ on leader and follower vehicles.

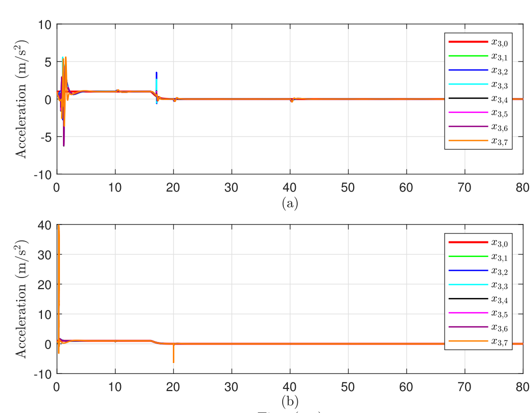

Fig. 4. Results of vehicular acceleration $x_{3,i}, i\in\{0,1,\ldots,7\}$ on leader and follower vehicles. (a) Proposed adaptive fuzzy AFTC. (b) Method in [40].

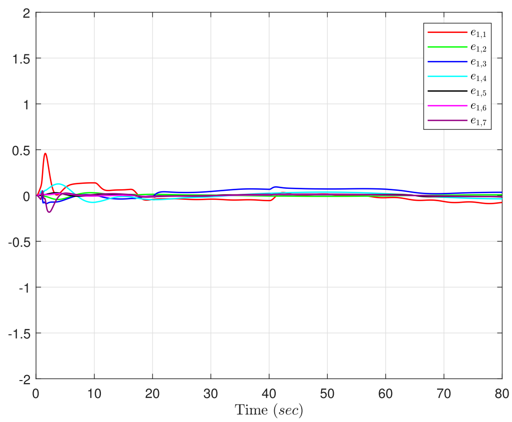

Fig. 5. Vehicle spacing errors between adjacent vehicles, where e_{1,i}, i = 1,...,7, denotes the spacing error between the (i-1)-th and i-th vehicles.

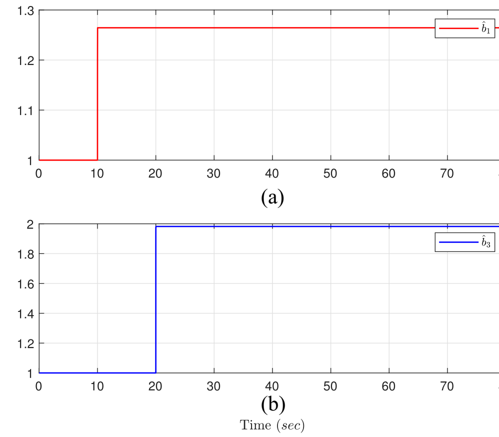

Fig. 8. Compensation coefficient for the fault factor of the ranging sensors on vehicles 1 and 3.

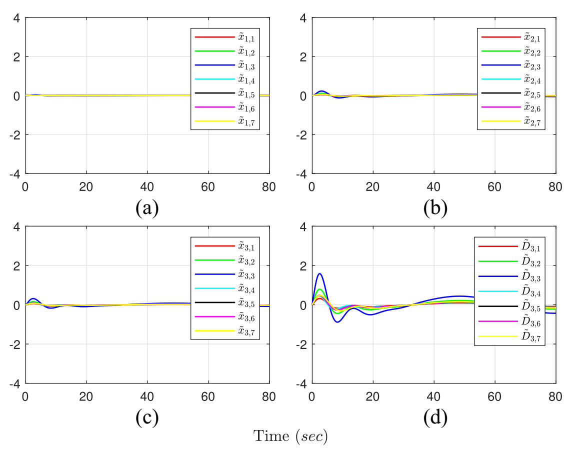

Fig. 6. Observation errors. (a) Position estimation error $x_{1,i} - \hat{x}_{1,i}$. (b) Velocity estimation error $x_{2,i} - \hat{x}_{2,i}$. (c) Acceleration estimation error $x_{3,i} - \hat{x}_{3,i}$. (d) Disturbance estimation error $D_i - \hat{D}_i$.

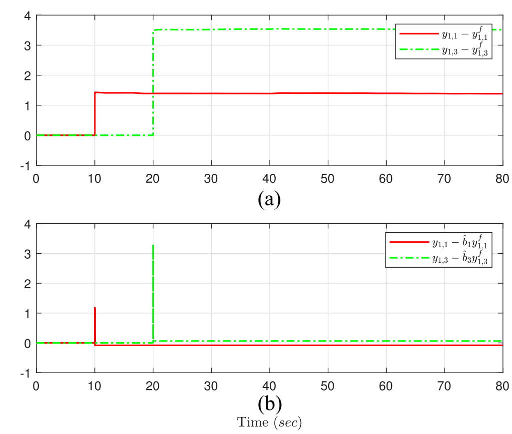

Fig. 9. Effect of the proposed compensation mechanism. (a) Uncompensated measurement error $y_{1,i} - y^f_{1,i}$. (b) Residual error $y_{1,i} - \hat{b}_i y^f_{1,i}$ after applying the compensation coefficient $\hat{b}_i$.

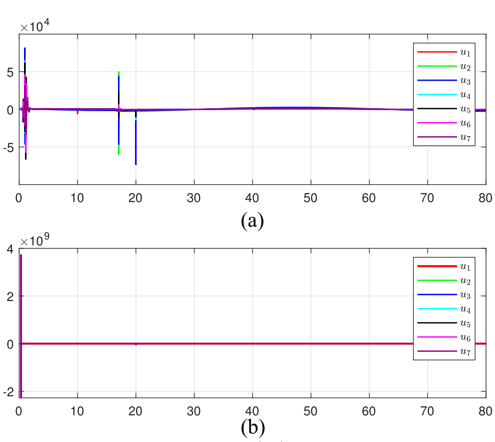

Fig. 7. Control input trajectories. (a) Proposed adaptive fuzzy AFTC. (b) Method in [40].

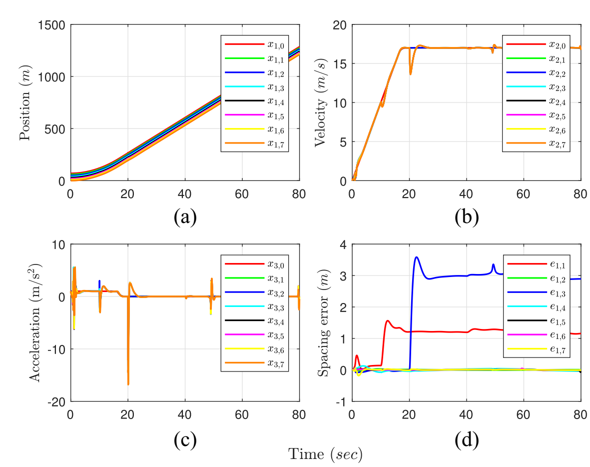

Fig. 10. Results of the vehicular platoon without fault compensation mechanism. (a) Position. (b) Velocity. (c) Acceleration. (d) Vehicle spacing errors between adjacent vehicles.

oscillations in vehicle acceleration during the initial transient, which is unfavorable for real-vehicle implementation and may impose undesirable mechanical stress on the actuators. In contrast, when the saturated Nussbaum function is adopted in the proposed adaptive fuzzy AFTC scheme, such excessive transient oscillations are effectively suppressed, and the acceleration responses become much smoother. These results further demonstrate the effectiveness and practical advantages of the proposed control strategy.

As shown in Fig. 5, the spacing errors of all follower vehicles remain uniformly bounded, which is consistent with the theoretical result and confirms that the proposed adaptive fuzzy AFTC guarantees string stability of the vehicular platoon in the sense of Definition 1.

Observation errors are depicted in Fig. 6. As shown in Fig. 6(a)–(c), the estimated trajectories of vehicles 1–7 accurately track their actual states, confirming the effectiveness and robustness of the proposed extended state fuzzy observer. Fig. 6(d) further illustrates the disturbance estimation error $D_i - \hat{D}_i$, indicating that the external disturbance is accurately reconstructed and that reliable disturbance information is provided to the controller. The corresponding control input trajectories are shown in Fig. 7(a), from which it can be observed that all control signals remain bounded. In contrast, the control input trajectories generated by the method in [40], as shown in Fig. 7(b), exhibit large-amplitude oscillations. This further highlights the superiority of the proposed method in achieving smoother control behavior, which is more favorable for practical implementation in real vehicular systems.

To illustrate the effectiveness of the proposed compensation mechanism in correcting faulty inter-vehicle distance information, Figs. 8–10 are presented. As shown in Fig. 8, when the ranging sensor faults occur as specified in Table II, the compensation mechanism is activated to correct the faulty inter-vehicle distance information. In Fig. 9(a), in the absence of the compensation mechanism, the vehicles operate on erroneous ranging measurements. By contrast, Fig. 9(b) depicts the residual error between the true inter-vehicle distance and the compensated measurement under the proposed scheme, showing that the adaptive compensation mechanism promptly adjusts the fault coefficients and maintains reliable relative distance. Fig. 10 shows the responses of the vehicular platoon in the absence of the compensation mechanism. As shown in Fig. 10(a)–(c), the vehicles exhibit improper motion responses, while Fig. 10(d) shows large tracking errors. These results demonstrate the severe impact of ranging sensor faults on platoon safety and stability, and further verify the effectiveness of the proposed compensation scheme.

VI. Conclusion

This article has investigated an adaptive fuzzy AFTC scheme for third-order nonlinear vehicular platoons. Unlike many existing studies that either consider only a single fault type or assume that inter-vehicle distance measurements are always reliable, this work develops an AFTC framework that simultaneously handles actuator loss-of-effectiveness faults and ranging sensor faults. Specifically, a saturated Nussbaum function is adopted to cope with unknown actuator fault gains, an adaptive compensation mechanism is designed to correct distorted inter-vehicle distance measurements, and an extended state fuzzy observer is constructed to estimate both nonlinear vehicle dynamics and external disturbances. Under the proposed design, internal stability and string stability of the vehicular platoon have been established via Lyapunov analysis and verified by numerical simulations. In addition, traffic flow stability is ensured under the adopted CD policy. However, since vehicular platoons rely on ranging sensor and inter-vehicle communications, potential network attacks on communication channels have not been considered in this study and will be explored in future work.

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